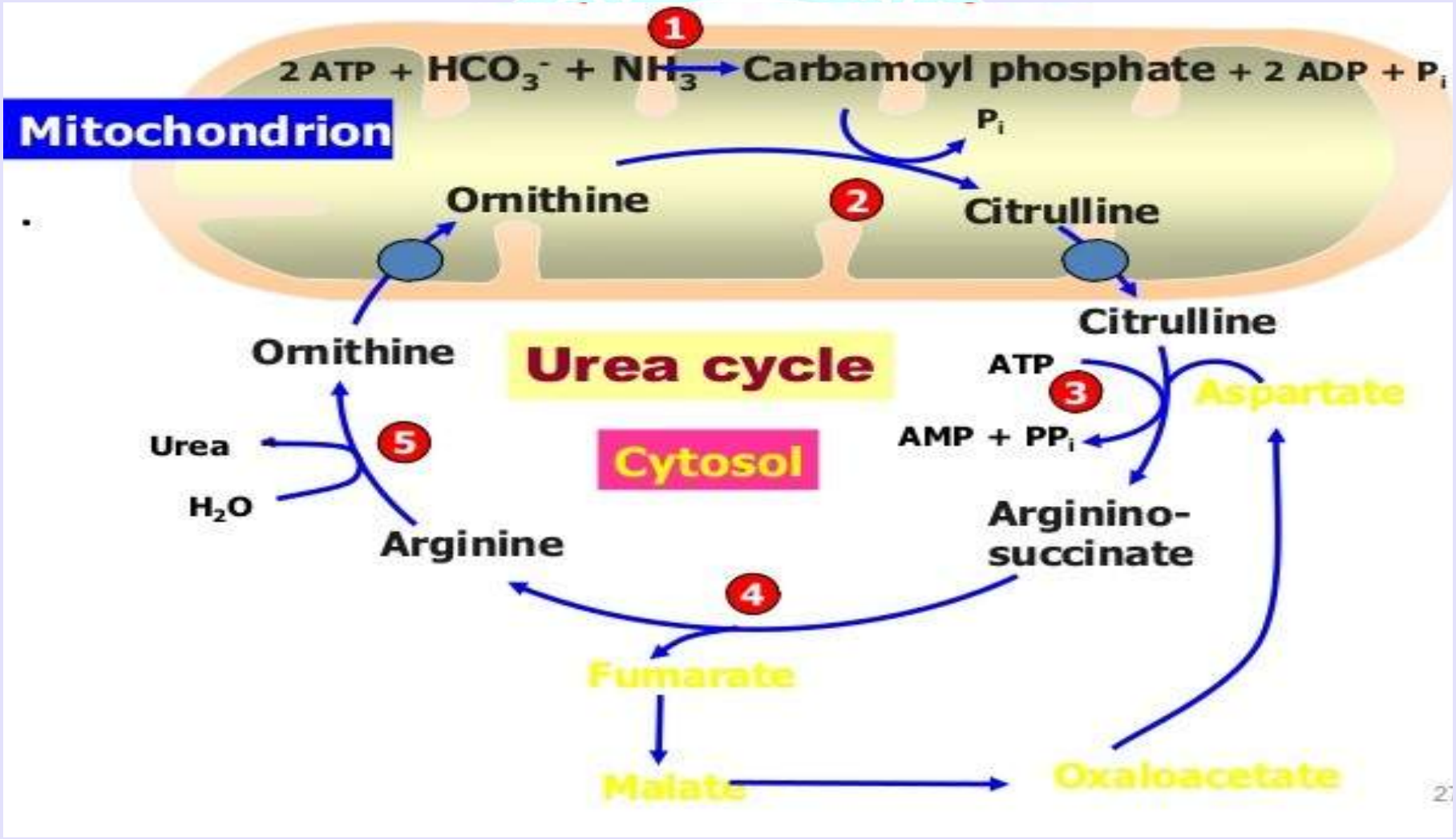
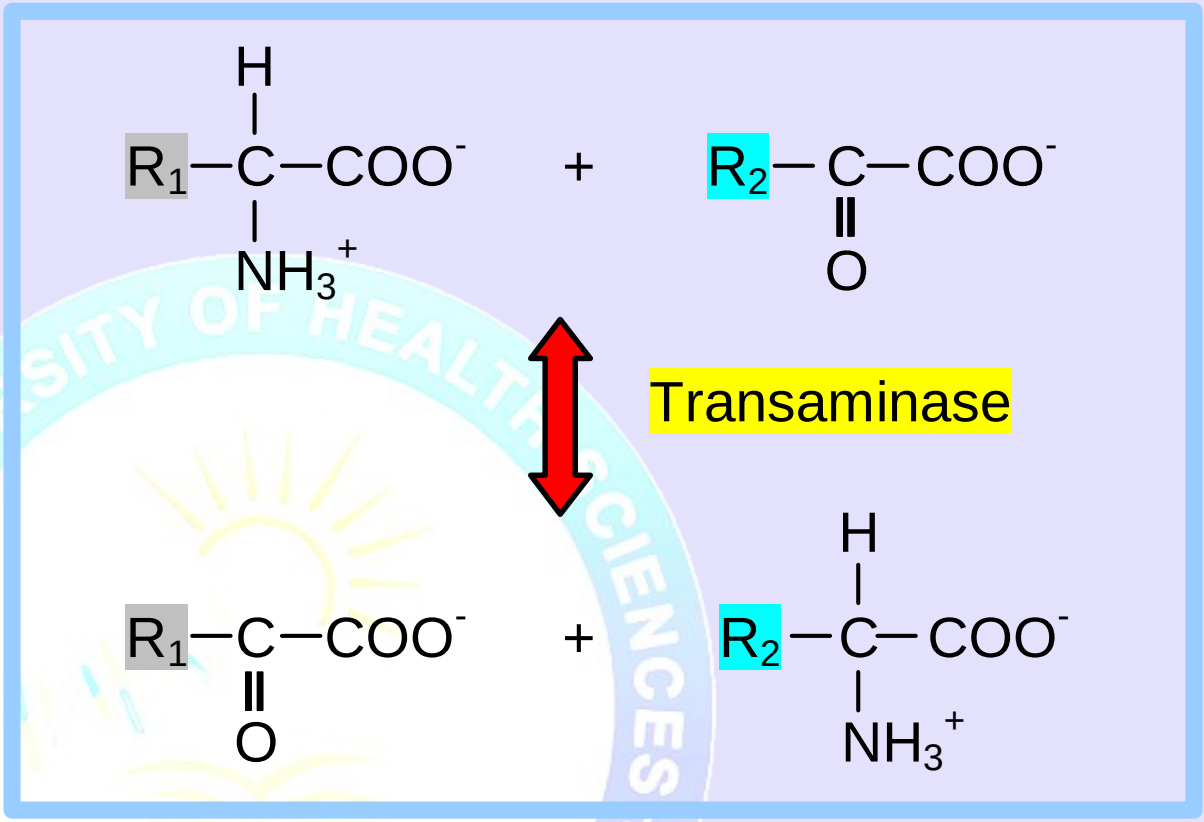
Nitrogen Metabolism, Ammonia Degradation and Urea Formation



OBJECTIVES

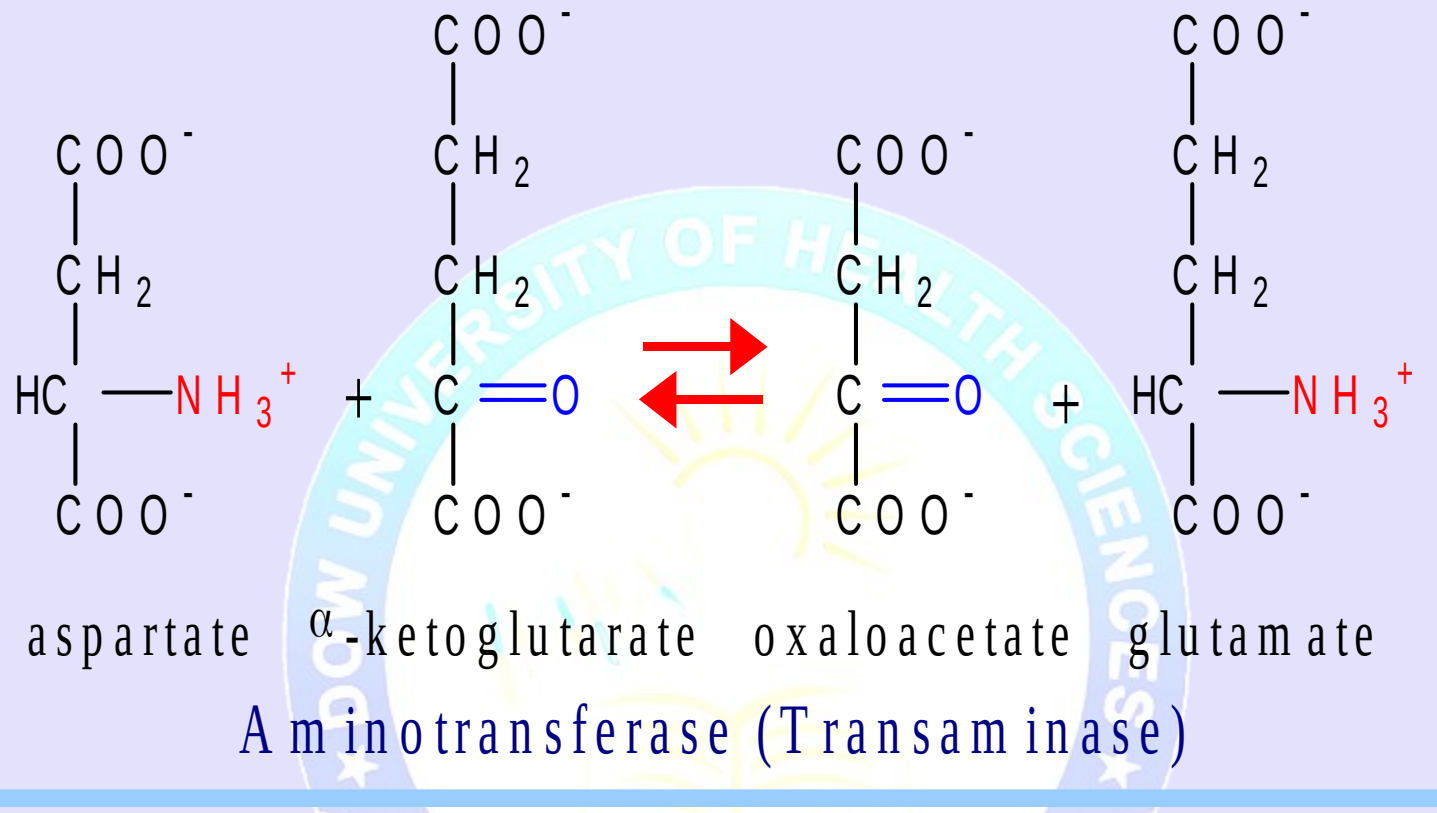
- Describe Nitrogen balance.
- Explain Negative and positive nitrogen balance.
- Discuss the reactions related with the degradation of amino acids.
- Explain ammonia formation and degradation.
- Discuss Urea formation.
- Enlist different steps of urea formation.

Transaminases (aminotransferases) catalyze the reversible reaction at right.



There are multiple transaminase enzymes which **vary in substrate specificity**.

Some show preference for particular **amino acids** or classes of amino acids as amino group donors, and/or for particular **α-keto acid acceptors**.



Example of a **Transaminase** reaction:

- Aspartate donates its amino group, becoming the α-keto acid oxaloacetate.
- α-Ketoglutarate accepts the amino group, becoming the amino acid glutamate.

Sources of Ammonia

1. Many tissues, particularly **liver**, form ammonia from amino acids by transamination and oxidative deamination
2. **Intestinal mucosa** hydrolysis glutamine by intestinal glutaminase



Sources of Ammonia

3. **The intestinal lumen** by the bacterial degradation of urea by <u>urease</u>



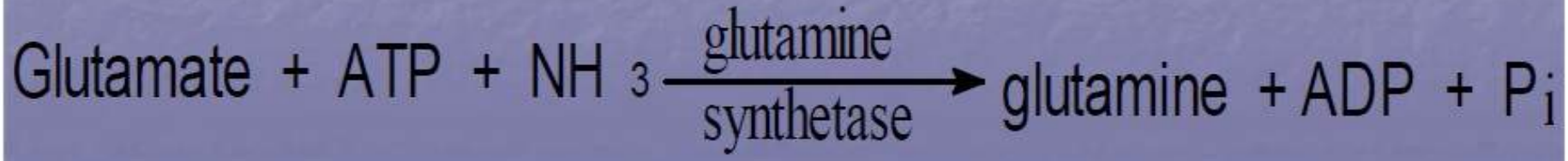
- Ammonia absorbed by the portal vein and is almost quantitatively removed by the liver

4. **Kidneys** degrades glutamine by <u>renal glutaminase</u> into ammonia & glutamte

- Most of this ammonia is excreted into the urine as NH_4^+

Transport of Ammonia in the Circulation

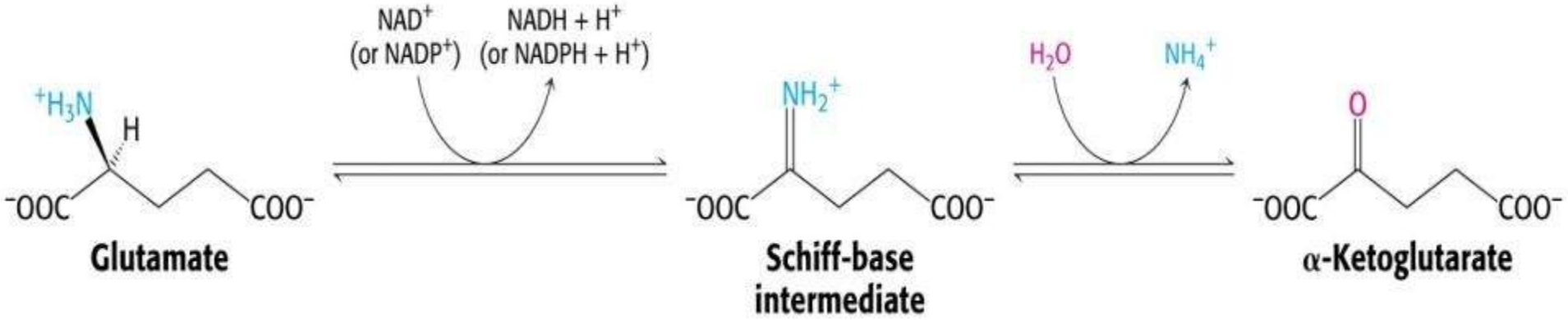
- Although ammonia is constantly produced in the tissues, it is present at very low levels in blood
- This is due to rapid removal of ammonia from blood by liver as well as many tissues, particularly muscle, in the form of **glutamine** or **alanine**, rather than as free ammonia



AMMONIA METABOLISM

The ways of ammonia formation

1. Oxidative deamination of amino acids



2. Deamination of physiologically active amines and nitrogenous bases.

3. Absorption of ammonia from intestine (degradation of proteins by intestinal microorganisms results in the ammonia formation).

4. Hydrolytic deamination of AMP in the brain (enzyme – *adenosine deaminase*)

Ammonia is a toxic substance to plants and animals (especially for brain)

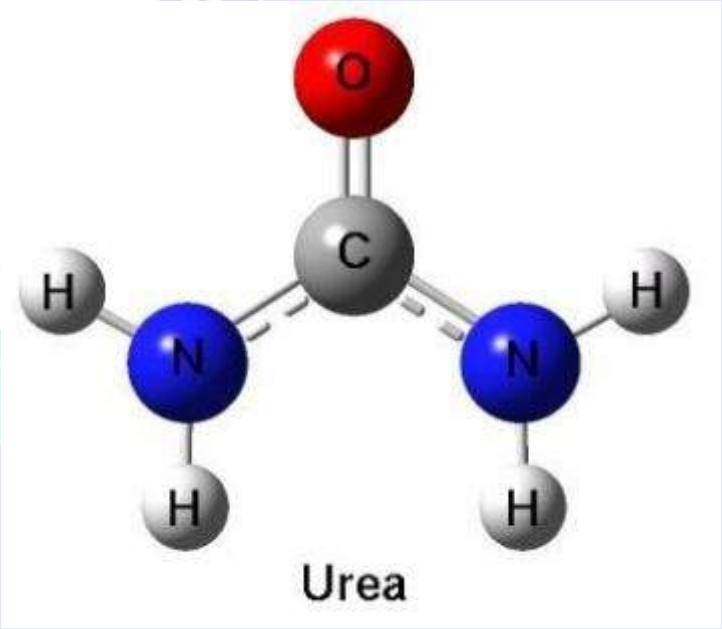
Normal concentration: **25-40 μmol/l** (0.4-0.7 mg/l)

Ammonia must be removed from the organism

Terrestrial vertebrates synthesize **urea** (excreted by the **kidneys**) - ureotelic organisms

Birds, reptiles synthesize **uric acid**

Urea formation takes place in the liver

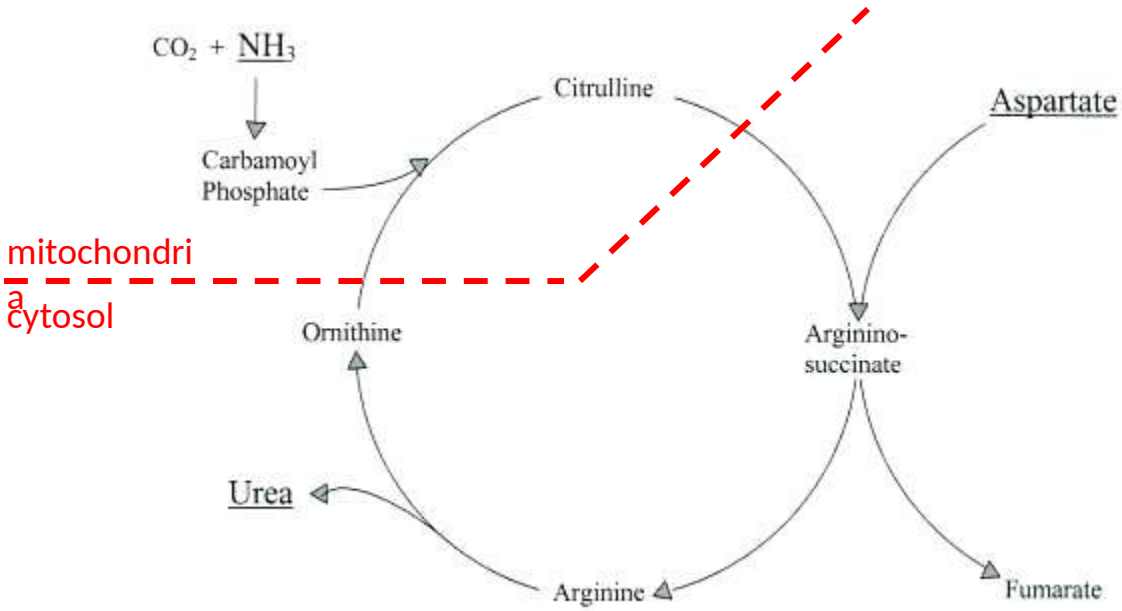


Urea

UREA

- Urea is the major disposal form of amino groups derived from amino acids.
- 90% of the nitrogen containing components of urine are urea.
- The carbon and oxygen of urea are derived from $CO_{2.}$
- Urea is produced by the liver, transported in the blood to the kidneys for excretion.

UREA CYCLE



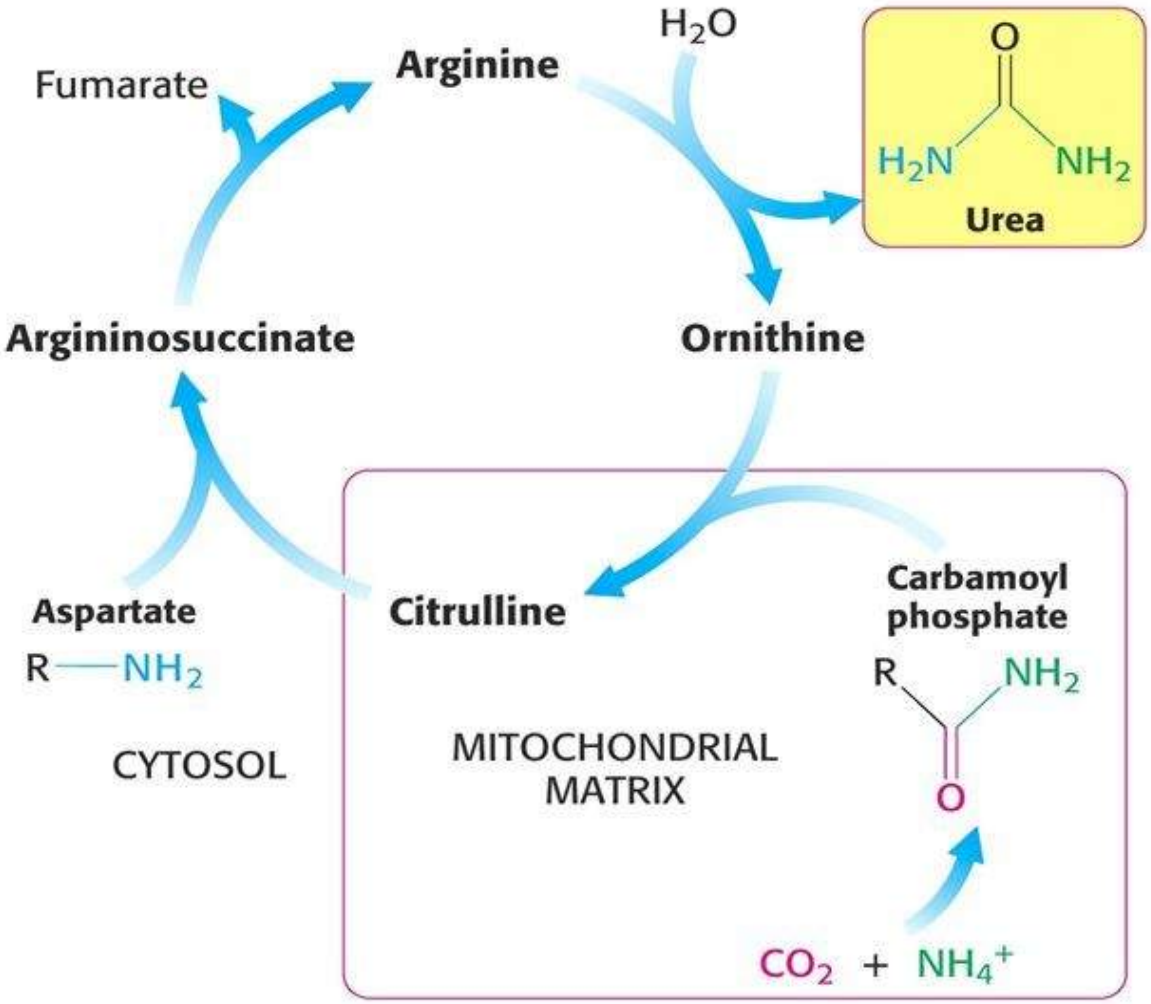
Function: detoxification of ammonia
(prevents hyperammonemia)

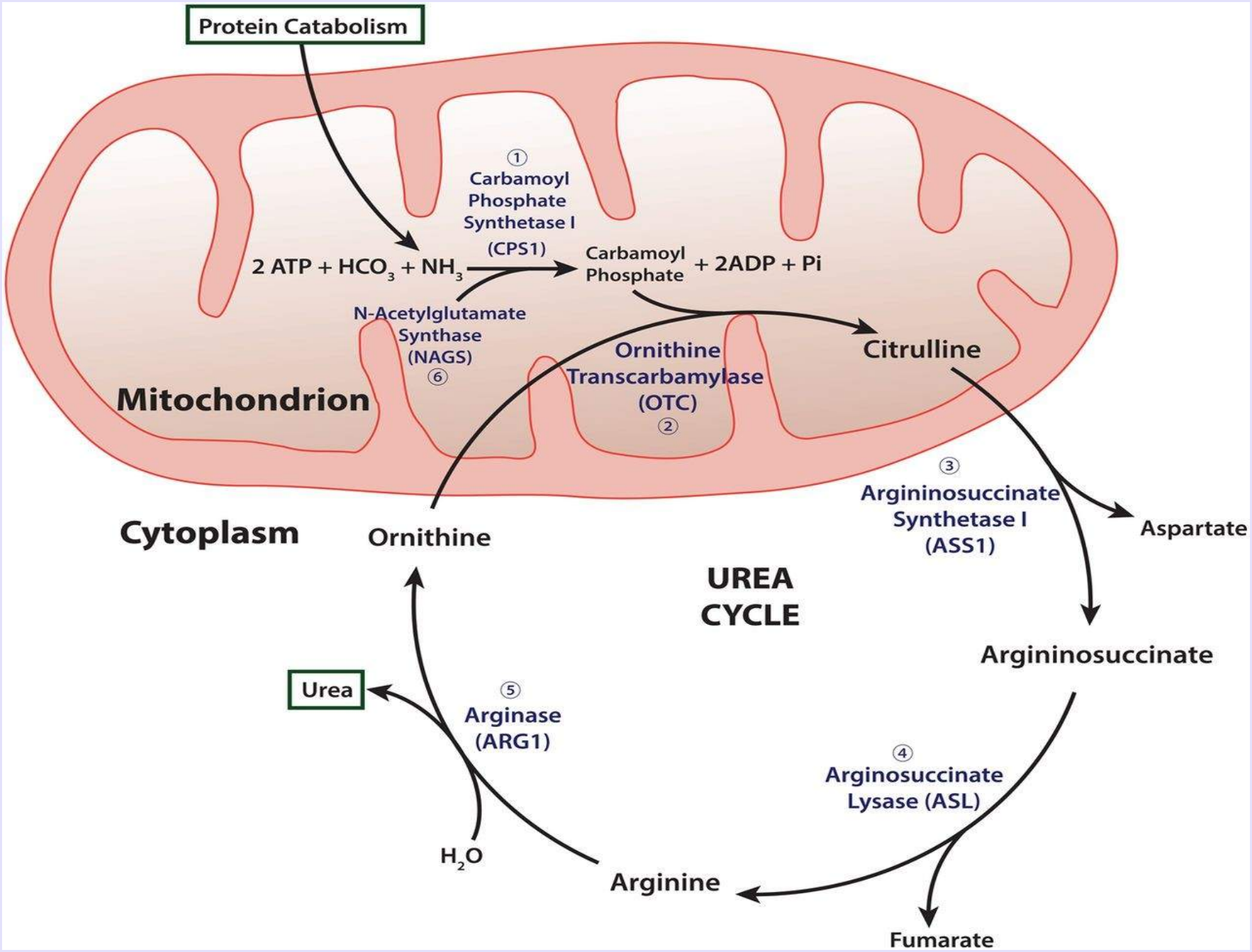
THE UREA CYCLE

Urea cycle - a cyclic pathway of urea synthesis first postulated by **H.Krebs**

The sources of nitrogen atoms in urea molecule:
- aspartate;
- NH_4^+.

Carbon atom comes from CO_2.




Protein Catabolism
1
Carbamoyl Phosphate Synthetase I (CPS1)
2 ATP + HCO3 + NH3
Carbamoyl Phosphate
+ 2ADP + Pi
N-Acetylglutamate Synthase (NAGS)
6
Mitochondrion
Ornithine Transcarbamylase (OTC)
2
Citrulline
3
Argininosuccinate Synthetase I (ASS1)
Aspartate
Cytoplasm
Ornithine
UREA CYCLE
Argininosuccinate
Urea
5
Arginase (ARG1)
4
Arginosuccinate Lysase (ASL)
H2O
Arginine
Fumarate

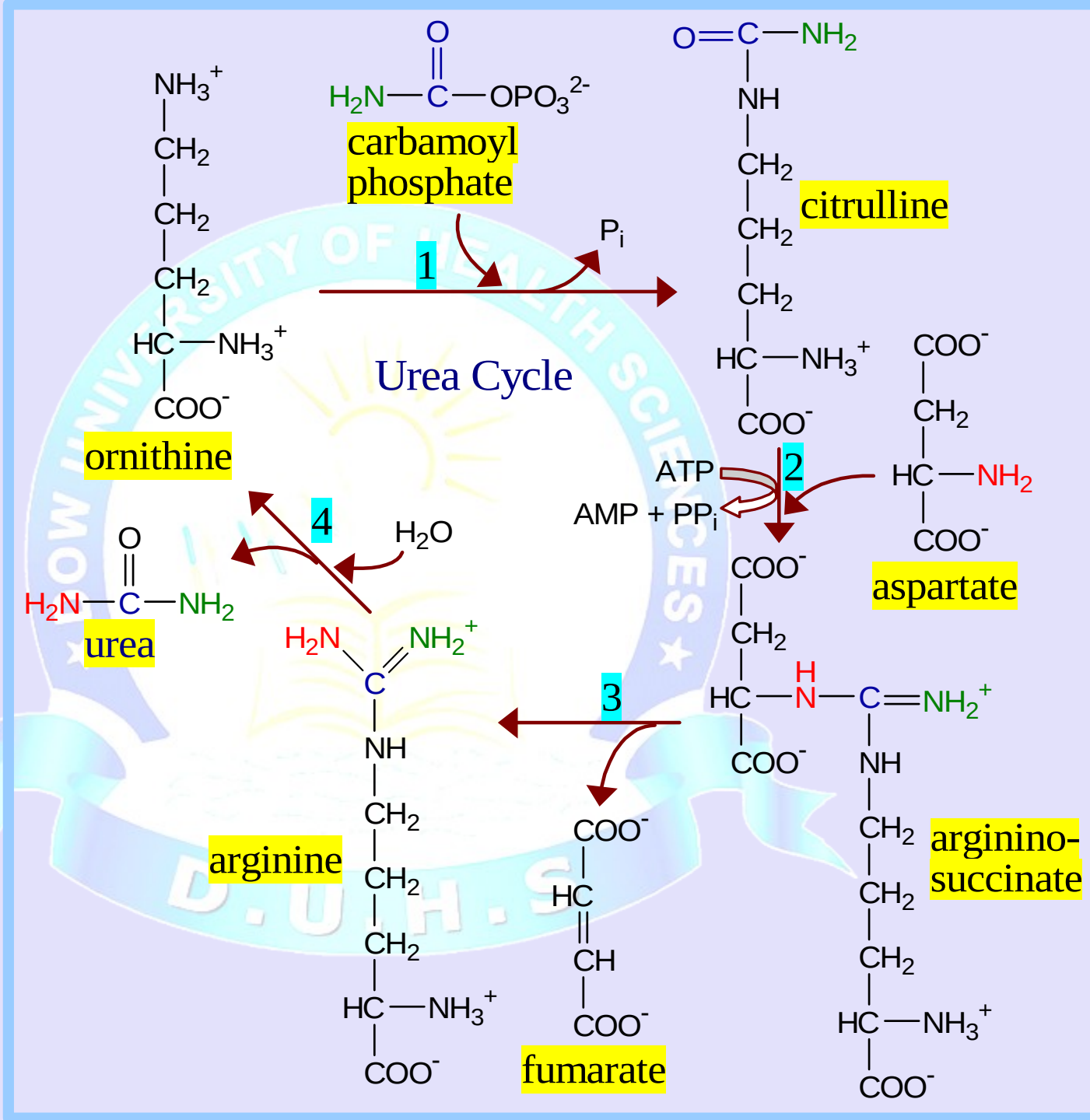
Urea Cycle

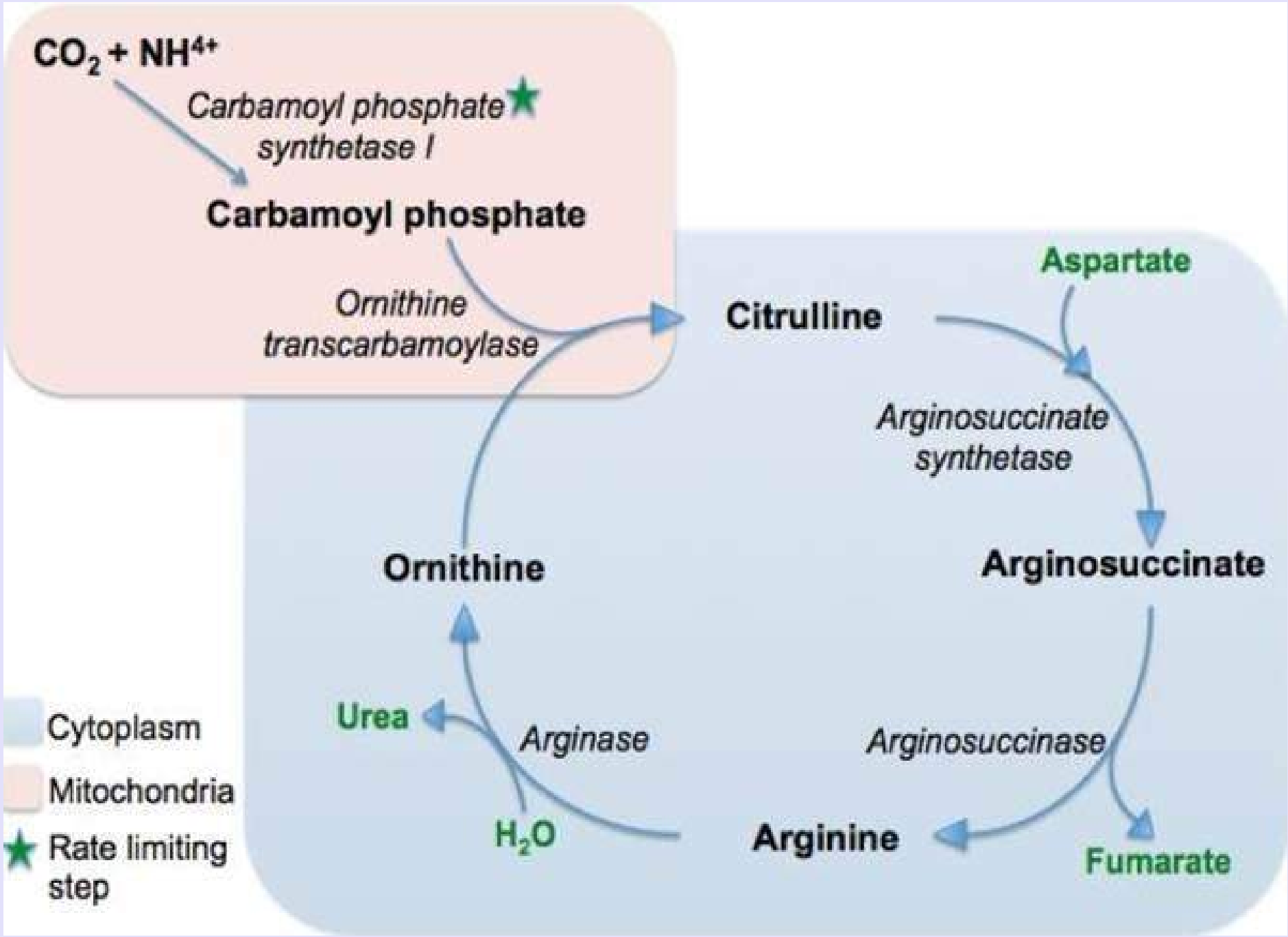
Enzymes in **mitochondria**:

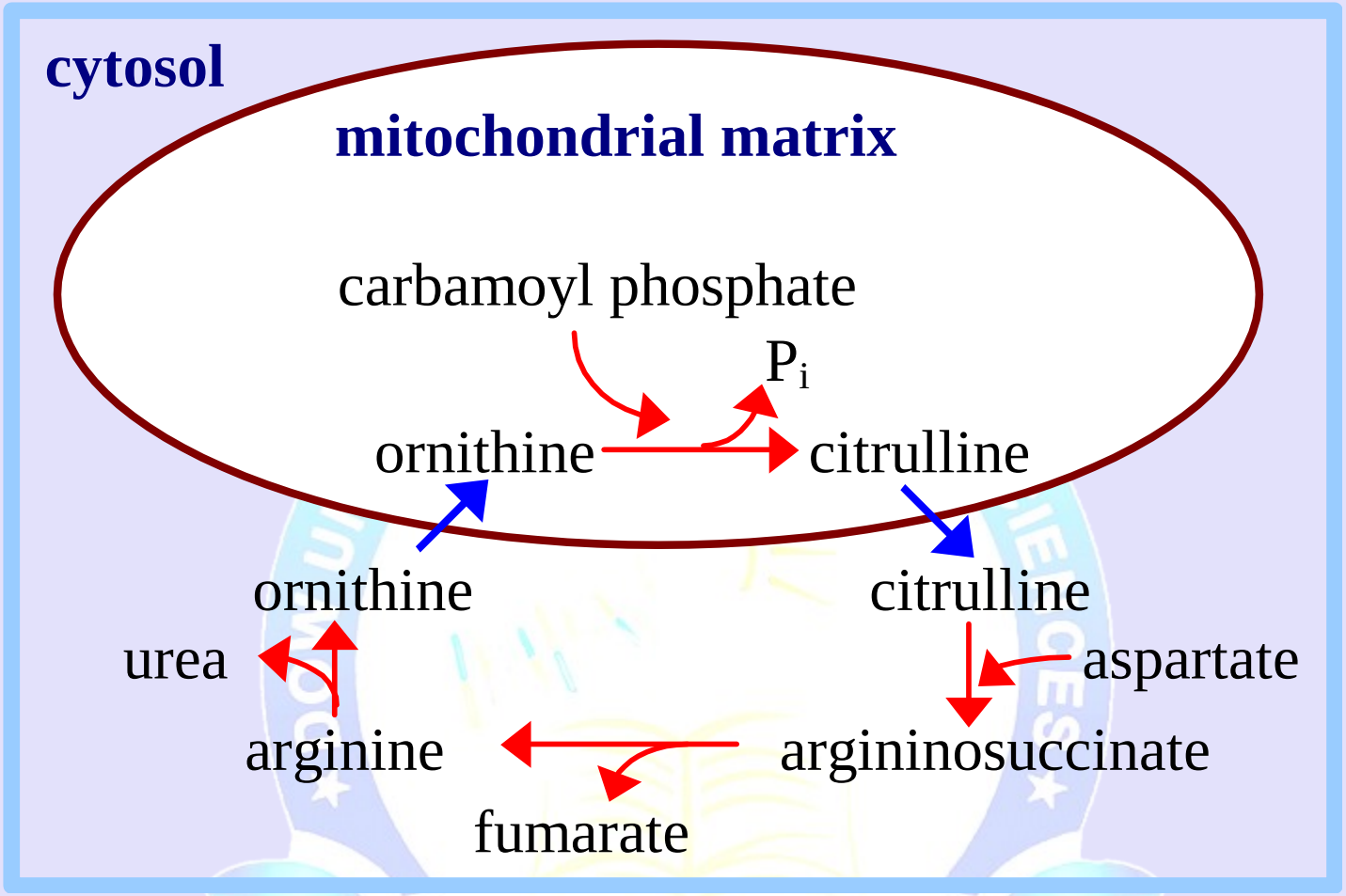
1. Ornithine Trans-carbamylase

Enzymes in **cytosol**:

2. Arginino-Succinate Synthase
3. Arginino-succinase
4. Arginase.

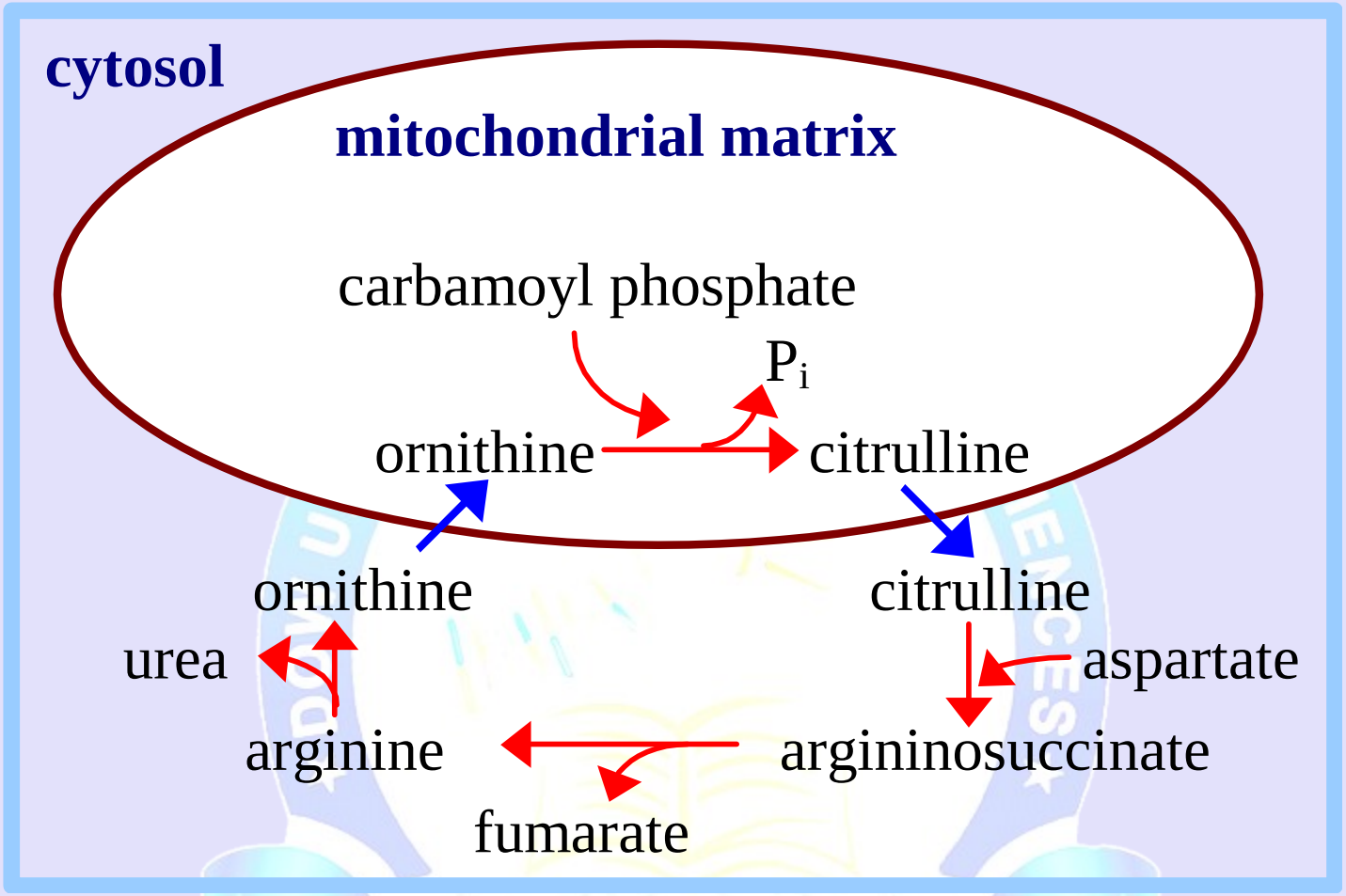



$CO_2 + NH^{4+}$
Carbamoyl phosphate synthetase I
Carbamoyl phosphate
Ornithine transcarbamoylase
Citrulline
Aspartate
Arginosuccinate synthetase
Arginosuccinate
Arginosuccinase
Fumarate
Arginine
H_2O
Arginase
Urea
Ornithine
Cytoplasm
Mitochondria
Rate limiting step



For each cycle, **citrulline** must leave the mitochondria, and **ornithine** must enter the mitochondrial matrix.

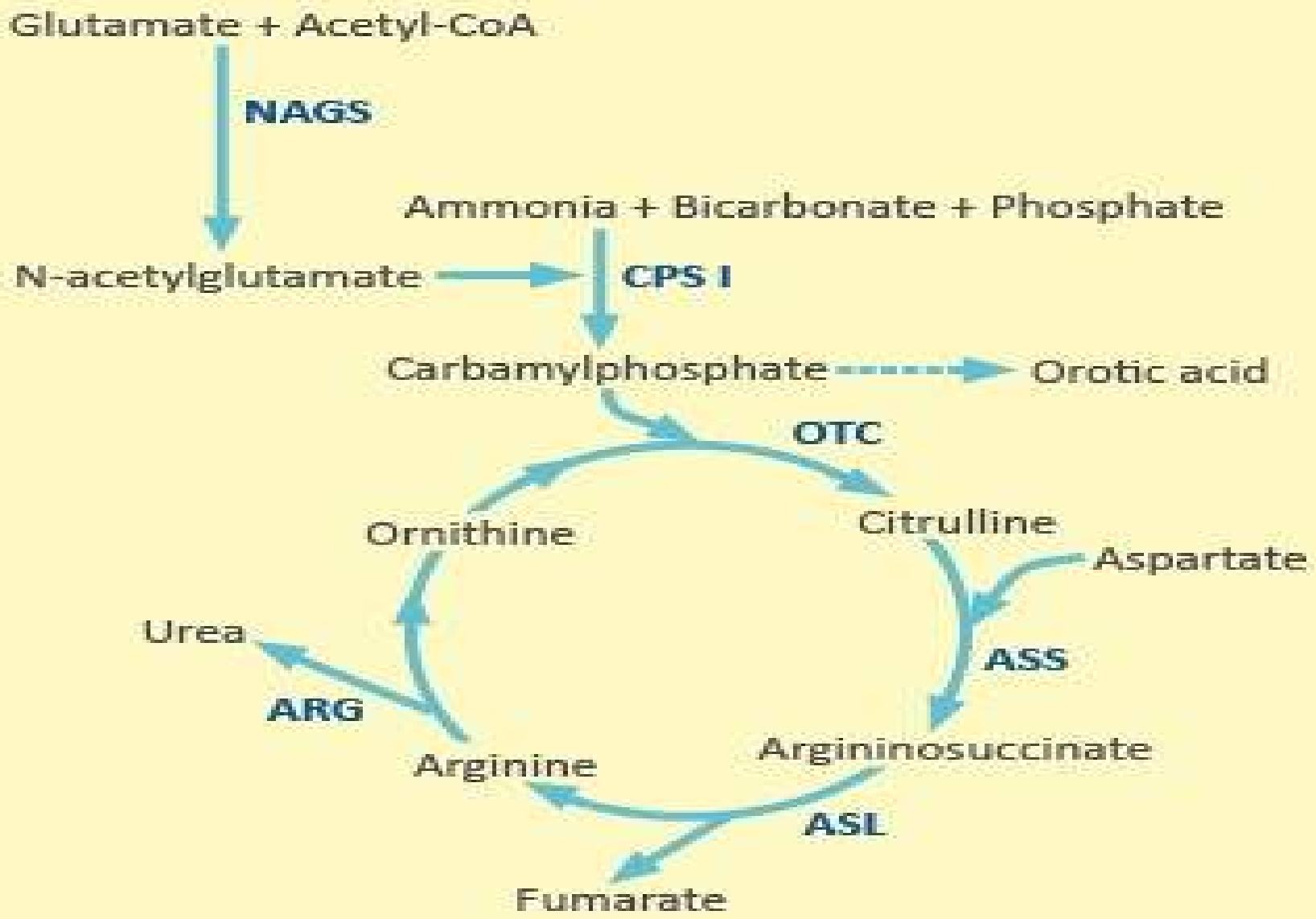
An **ornithine/citrulline transporter** in the inner mitochondrial membrane facilitates transmembrane fluxes of citrulline & ornithine.

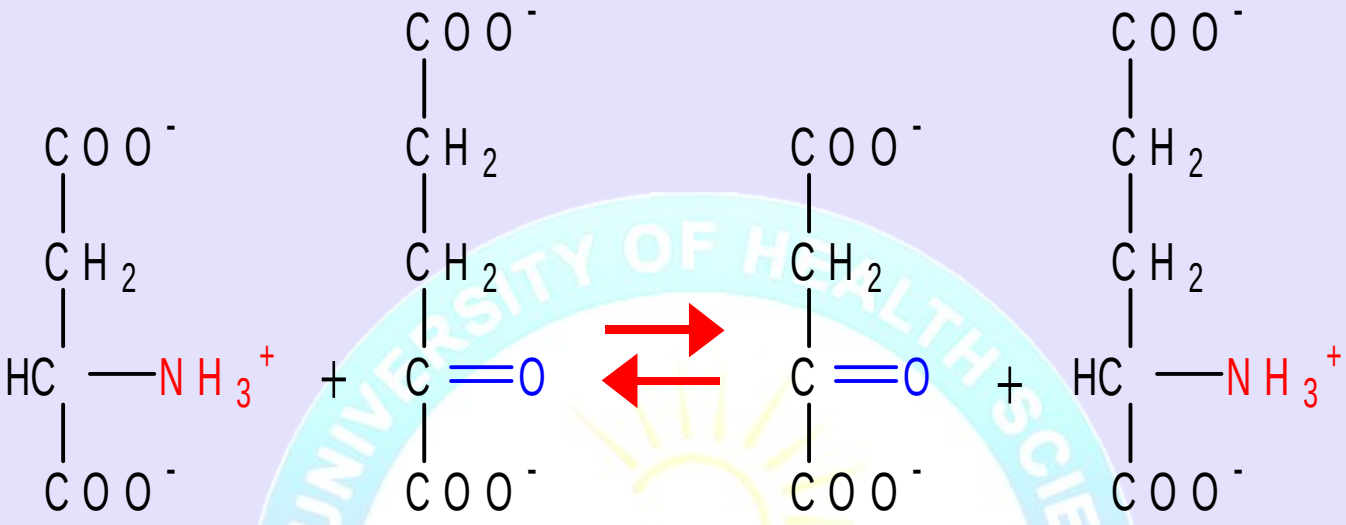


A complete **Krebs Cycle** functions only within mitochondria.

But cytosolic isozymes of some Krebs Cycle enzymes are involved in regenerating **aspartate** from **fumarate**.

The Urea Cycle



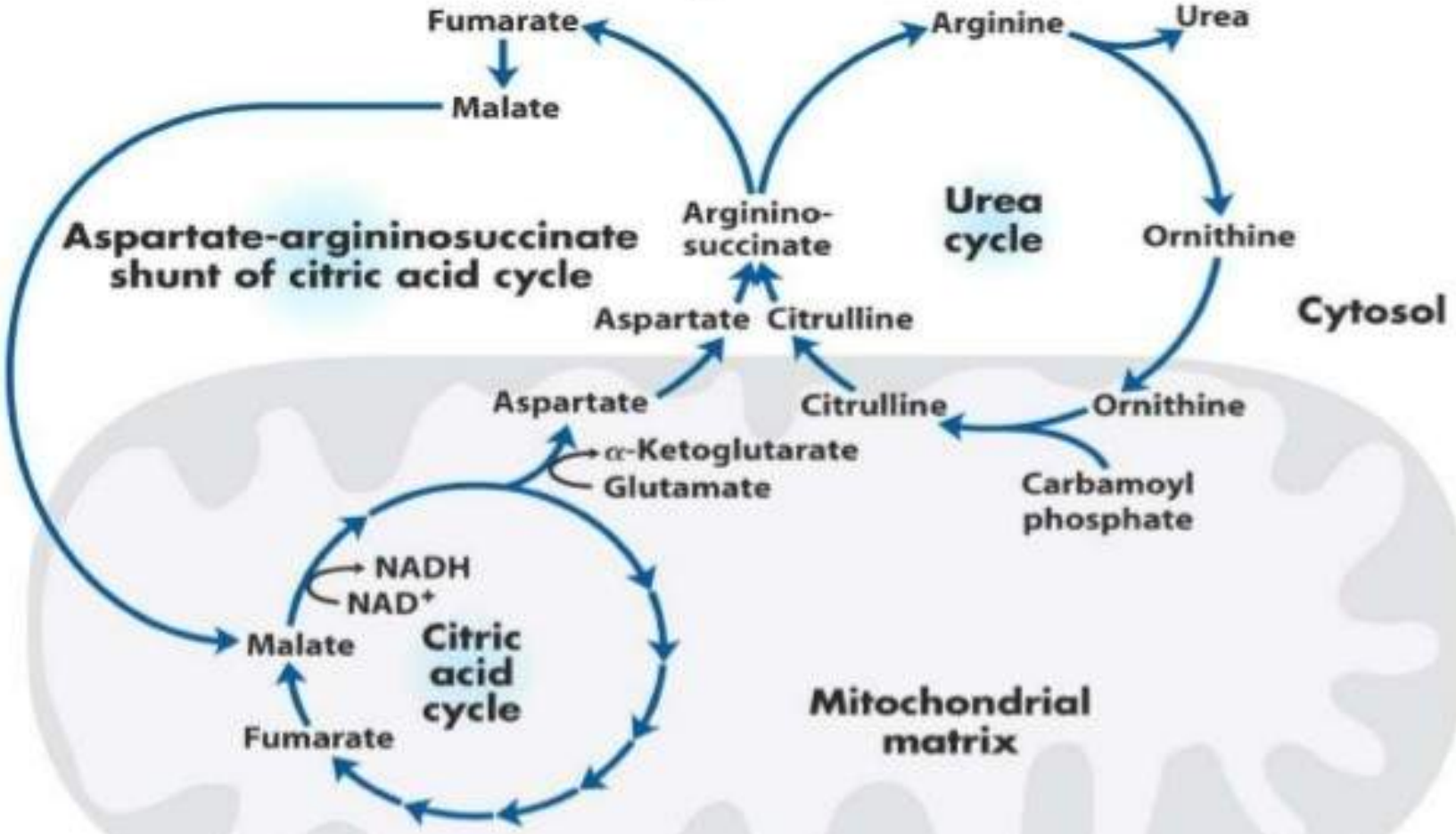


Fumarate is converted to **oxaloacetate** via Krebs Cycle enzymes Fumarase & Malate Dehydrogenase.

Oxaloacetate is converted to **aspartate** via transamination (e.g., from glutamate).

Aspartate then reenters Urea Cycle, carrying an amino group derived from another amino acid.

Relationship of Urea cycle and TCA cycle



Summary of Amino Acid Catabolism

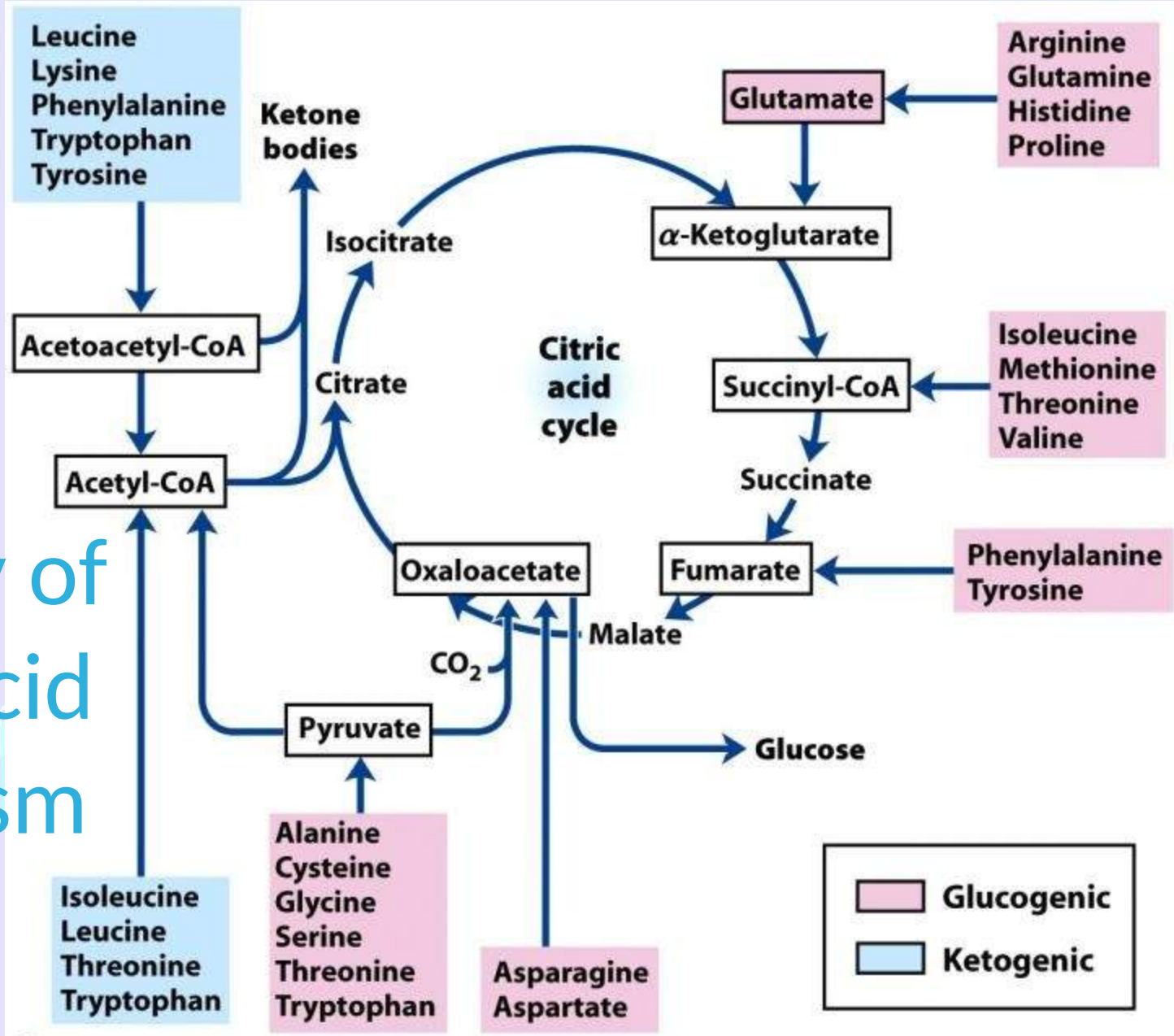


Figure 18-15
Lehninger Principles of Biochemistry, Fifth Edition
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Hereditary deficiency of any of the Urea Cycle enzymes leads to **hyperammonemia** - elevated [ammonia] in blood.

Total lack of any Urea Cycle enzyme is lethal.

Elevated ammonia is toxic, especially to the brain.

If not treated immediately after birth, severe mental retardation results.

Urea cycle disorders

Carbamoyl Phosphate synthetase (CPS-1) deficiency

- Along with OTC deficiency, deficiency of CPS-I is the most severe of the urea cycle disorders.
- Defects in the enzyme carbamoyl phosphate synthase I are responsible for the relatively rare (estimated frequency 1:62,000) metabolic disease termed "hyperammonemia type 1."
- Individuals with complete CPS-I deficiency rapidly develop hyperammonemia in the newborn period.
- Children who are successfully rescued from crisis are chronically at risk for repeated bouts of hyperammonemia

Deficiency	Disorder	Clinical Feature
N-Acetylglutamate synthase	Hyperammonemia that may be accompanied by high plasma concentrations of alanine and glutamine	Lethargy; persistent vomiting; poor feeding; hyperventilation; enlarged liver; seizures
Carbamoyl phosphate synthetase	Hyperammonemia; citrullinemia; respiratory alkalosis	Lethargy; coma; seizures; vomiting; poor feeding; hyperventilation; hepatomegaly
Ornithine transcarbamylase	Hyperammonemia; respiratory alkalosis; elevated orotic acid in urine	Seizures; vomiting; poor feeding; hyperventilation; hepatomegaly
Arginosuccinate synthetase	Citrullinemia	Lethargy; coma; seizures; vomiting; poor feeding; hepatomegaly
Arginosuccinate lyase	Elevated arginosuccinic acid in urine	Lethargy; seizures; vomiting; poor feeding; hyperventilation; hepatomegaly
Arginase	Markedly elevated plasma arginine, lactate, and CSF glutamine, and modestly elevated blood ammonia	Delayed development; protein intolerance; spasticity; loss of muscle control; seizures; irritability

CSF indicates cerebrospinal fluid.



Thank you